Refugee SchoolOS

Wireframes, Build Phases, and Codex Prompt Pack

A SaaS-ready LMS and school management platform for refugee learning centres

Purpose: Build a practical LMS for refugee schools that combines school records, attendance, LMS, e-library, video lessons, reports, and refugee support workflows.

Primary users: Super Admin, School Admin, Teacher, Student, Guardian/Parent, and optional Case Manager.

Recommended stack: Next.js, TypeScript, TailwindCSS, PostgreSQL, Prisma, Cloudflare R2/DigitalOcean Spaces, Playwright/Puppeteer PDF, Brevo, and Twilio/WhatsApp Cloud API.

Key SaaS rule: every major data table must include school_id for tenant isolation.

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1. Product Scope

Refugee SchoolOS should be designed as an LMS plus school management platform. It should not only deliver lessons; it should also help refugee schools manage students, classes, attendance, grades, records, documents, payments, and support cases.

Module	Core Features
Student Records	Profiles, photos, class, guardian, emergency contacts, optional refugee document fields, soft delete
Attendance	Daily attendance, late/absent reasons, class/date filtering, monthly reports, PDF/Excel export
Classes & Subjects	Pre-GED, GED, language classes, custom classes, teacher assignment, timetables
LMS	Lessons, assignments, submissions, quizzes, teacher grading, student portal
E-Library	Books, PDFs, categories, subjects, reading levels, language filters, cover images, downloads
Video Lessons	Class-based lessons, YouTube/Vimeo/private links, playlists, watch progress, teacher analytics
Exams & Grades	GED subjects, custom exams, marks entry, report cards, progress summaries
Reports	Attendance reports, student progress, class performance, finance/receipt reports, export tools
Refugee Support	Case notes, document expiry reminders, sponsor support, sensitive permissions, audit logs
School Branding	Custom school name, logo, colors, contact, custom dashboard title, PDF branding
SaaS Admin	Multi-school onboarding, school settings, super admin dashboard, tenant isolation

2. Navigation and Information Architecture

```
Public Website
├── Landing Page
├── Features
├── Pricing / NGO Plan
├── Request Demo
└── School Login

Dashboard
├── Overview
├── Students
├── Classes
├── Attendance
├── E-Library
├── Video Lessons
├── Lessons / LMS
├── Assignments
├── Exams & Grades
├── Payments / Receipts
├── Refugee Support
├── Reports
└── Settings
    ├── School Branding
    ├── Users & Roles
    ├── Subjects
    └── Data Export / Backup
```

3. Wireframes

3.1 Public Landing Page

```
┌──────────────────────────────────────────────────────────────┐
│ Refugee SchoolOS        Features  E-Library  Pricing  Login  │
├──────────────────────────────────────────────────────────────┤
│ HERO                                                          │
│ Manage refugee schools with one simple LMS and school system. │
│ [Start Free] [Request Demo] [School Login]                    │
│                                                               │
│ Dashboard screenshot mockup / product preview                 │
├──────────────────────────────────────────────────────────────┤
│ FEATURE CARDS                                                 │
│ Student Records | Attendance | LMS | E-Library | Video Lessons │
├──────────────────────────────────────────────────────────────┤
│ BUILT FOR REFUGEE SCHOOLS                                     │
│ Low cost | Multilingual | Printable reports | Sensitive data  │
├──────────────────────────────────────────────────────────────┤
│ E-LIBRARY + VIDEO LESSON HIGHLIGHT                            │
│ Upload books, PDFs, and video classes for students             │
├──────────────────────────────────────────────────────────────┤
│ CUSTOM SCHOOL BRANDING                                        │
│ Each school can use its own name, logo, colors, and reports    │
├──────────────────────────────────────────────────────────────┤
│ PRICING / NGO PLAN                                            │
│ Free NGO starter plan | Basic | Pro | Network                  │
├──────────────────────────────────────────────────────────────┤
│ FAQ + CTA + FOOTER                                            │
└──────────────────────────────────────────────────────────────┘
```

3.2 School Dashboard

```
┌──────────────────────────────────────────────────────────────┐
│ [Logo] Mon Refugee Learning Centre        Search   [Profile] │
├────────────────┼─────────────────────────────────────────────┤
│ Sidebar        │ Welcome, MRLC Admin                         │
│                │                                             │
│ Dashboard      │ KPI Cards                                   │
│ Students       │ [Total Students] [Present Today] [Absent]   │
│ Classes        │ [Fees Due]       [Books]         [Videos]   │
│ Attendance     │                                             │
│ E-Library      │ Charts                                      │
│ Video Lessons  │ Attendance Trend | Grade Progress            │
│ Lessons        │                                             │
│ Assignments    │ Quick Actions                               │
│ Exams          │ + Add Student | Take Attendance | Upload Book │
│ Payments       │ + Add Video Lesson | Create Exam             │
│ Refugee Cases  │                                             │
│ Reports        │ Recent Activity                             │
│ Settings       │                                             │
└────────────────┴─────────────────────────────────────────────┘
```

3.3 Student Profile

```
Student Profile
┌──────────────────────────────────────────────────────────────┐
│ Photo | Name | Student ID | Class | Status                    │
├──────────────────────────────────────────────────────────────┤
│ Tabs: Overview | Attendance | Grades | Payments | Documents  │
│       Library Activity | Video Progress | Case Notes          │
├──────────────────────────────────────────────────────────────┤
│ Overview                                                      │
```

```
| - Basic info: gender, date of birth, nationality, language    |
| - Guardian and emergency contact                              |
| - Class/enrollment history                                    |
| - Optional refugee document status                            |
| - Recent attendance and grade summary                         |
└──────────────────────────────────────────────────────────────┘
```

3.4 Attendance

```
Attendance / GED Class / Today
┌──────────────────────────────────────────────────────────────┐
| Date: [2026-05-11]   Class: [GED]   [Load Students]          |
├──────────────────────────────────────────────────────────────┤
| Student Name        Status                          Note     |
| Aung Mon            [Present Late Absent Excused]  ________  |
| Mya Hnin            [Present Late Absent Excused]  ________  |
| Nai Htoo            [Present Late Absent Excused]  ________  |
├──────────────────────────────────────────────────────────────┤
| [Save Attendance] [Export PDF] [Export Excel]                |
└──────────────────────────────────────────────────────────────┘
```

3.5 E-Library

```
E-Library
┌──────────────────────────────────────────────────────────────┐
| Search books... [Subject] [Language] [Reading Level] [+ Book]|
├──────────────────────────────────────────────────────────────┤
| Book Card          Book Card          Book Card              |
| [Cover]            [Cover]            [Cover]                |
| GED Science        English Grammar    Mon Reading            |
| PDF | Grade GED    PDF | Pre-GED      PDF | Language         |
| [View] [Download]  [View] [Download]  [View] [Download]      |
├──────────────────────────────────────────────────────────────┤
| Teacher/Admin: upload book, cover image, category, subject   |
| Student: view/download only books from own school            |
└──────────────────────────────────────────────────────────────┘
```

3.6 Video Lessons

```
Video Lessons
┌──────────────────────────────────────────────────────────────┐
| Search videos... [Class] [Subject] [Status] [+ Video Lesson] |
├──────────────────────────────────────────────────────────────┤
| Playlist / Lesson List                Video Player           |
| 1. GED Math: Algebra Basics            ┌──────────────────┐  |
| 2. Science: Cell Structure             |                  |  |
| 3. English: Essay Writing              |    Video         |  |
|                                        └──────────────────┘  |
|                                        Progress: 68%         |
├──────────────────────────────────────────────────────────────┤
| Teacher Analytics: watched, completed, last watched          |
└──────────────────────────────────────────────────────────────┘
```

3.7 School Branding Settings

```
School Branding Settings
┌──────────────────────────────────────────────────────────────┐
| School Name:        [Mon Refugee Learning Centre           ] |
| Short Name:         [MRLC                                  ] |
| Logo:               [Upload Logo]                            |
| Primary Color:      [#0F4C75]                                |
| Secondary Color:    [#3282B8]                                |
```

```
│ Address:            [..................................... ] │
│ Phone / Email:      [..................................... ] │
│ Website:            [https://monrefugeelc.com              ] │
│ Subdomain:          [mrlc.refugeeschoolos.com              ] │
├──────────────────────────────────────────────────────────────┤
│ Live Preview: dashboard header, login page, PDF report header │
│ [Save Branding]                                               │
└──────────────────────────────────────────────────────────────┘
```

4. Development Phases

Phase	Goal	Deliverables
Phase 1 - Foundation	Create the app base and SaaS-ready database	Next.js app, Prisma/PostgreSQL, auth-ready models, roles, school_id tenant structure, dashboard shell
Phase 2 - School Management MVP	Make it usable by one school	Student records, classes, attendance, reports, PDF/Excel export
Phase 3 - LMS Core	Add digital learning workflows	Subjects, lessons, assignments, submissions, grading, student dashboard
Phase 4 - E-Library & Video Lessons	Add book and video learning content	E-library, uploads, filters, video lessons, playlists, progress tracking
Phase 5 - Refugee Support	Support refugee school realities	Case notes, document tracking, sponsor support, emergency contacts, sensitive permissions
Phase 6 - SaaS Multi-School	Allow many schools to use the system	Super admin, school onboarding, branding, tenant guards, optional subdomains
Phase 7 - Production Hardening	Prepare for real deployment	Audit logs, validation, rate limits, backups, tests, secure headers, deployment guide

5. Database and SaaS Architecture

For SaaS safety, every major school-owned record must include school_id. Access control must always check both the user role and the user school.

```
Core Tables
schools
users
roles
students
guardians
classes
subjects
enrollments
attendance
lessons
assignments
submissions
exams
grades
payments
receipts
case_notes
documents
library_books
video_lessons
video_progress
announcements
audit_logs

Important SaaS rule:
- Every major table includes school_id.
- Middleware and service functions must prevent cross-school access.
- Sensitive records such as case_notes and refugee document fields need extra permissions and audit logs.
```

6. Codex Prompt Pack

Use these prompts one at a time. After each prompt, test the app, commit the working version, then continue to the next prompt.

Prompt 1 - Project Foundation

```
You are a senior full-stack SaaS engineer.

Build the foundation for a refugee school LMS called Refugee SchoolOS.

Tech stack:
- Next.js App Router
- TypeScript
- TailwindCSS
- Prisma
- PostgreSQL
- Role-based access control
- Multi-tenant-ready database using school_id

Create:
1. Clean project structure
2. Prisma schema
3. Auth-ready user model
4. School model
5. Student model
6. Class model
7. Enrollment model
8. Attendance model
9. Basic dashboard layout
10. Sidebar navigation

Important:
- Every major table must include school_id.
- Use clean TypeScript types.
- Add seed data for one school: Mon Refugee Learning Centre.
- Add roles: SUPER_ADMIN, SCHOOL_ADMIN, TEACHER, STUDENT, CASE_MANAGER.
- Make UI mobile responsive.
- Include .env.example and README setup instructions.
```

Prompt 2 - Landing Page and Login

```
Build a beautiful SaaS landing page for Refugee SchoolOS at /.

Sections:
1. Header with logo, Features, E-Library, Pricing, Request Demo, School Login
2. Hero section with strong education/NGO message
3. Dashboard preview mockup
4. Feature cards: Student Records, Attendance, LMS, E-Library, Video Lessons, Reports
5. Refugee-school benefits: low cost, multilingual, printable reports, sensitive data control
6. E-library and video lesson highlight section
7. Custom school branding section
8. Pricing / NGO plan section
9. FAQ
10. Footer with contact, privacy, terms

Design requirements:
- Modern education SaaS style
- TailwindCSS
- Fully responsive
- Clear CTAs: Start Free, Request Demo, School Login
- Add /login page for school users
- Do not connect billing yet.
```

Prompt 3 - Student Records Module

```
Build the Student Records module for Refugee SchoolOS.

Features:
```

```
1. Student list page
2. Search and filter by class, gender, status
3. Add student form
4. Edit student form
5. Student profile page
6. Guardian information
7. Emergency contact
8. Optional refugee document fields:
   - UNHCR status
   - document type
   - document number
   - expiry date
9. Upload student photo
10. Soft delete student

Security:
- Teachers can view only students from assigned classes.
- School admins can manage all students in their school.
- Super admin can access all schools.
- Do not expose sensitive document fields unless user has permission.

UI:
- Use clean cards and tables.
- Make mobile layout usable.
```

Prompt 4 - Attendance Module

```
Build the Attendance module.

Features:
1. Select class and date
2. Show student list
3. Mark Present, Late, Absent, Excused
4. Add note per student
5. Save attendance in bulk
6. Prevent duplicate attendance for same student/date/class
7. Monthly attendance report
8. Export attendance to PDF and Excel
9. Dashboard card showing present/absent today

Database:
- attendance table must include school_id, class_id, student_id, date, status, note, recorded_by.

Permissions:
- Teachers can take attendance for assigned classes.
- School admins can edit attendance.
- Students can only view their own attendance.
```

Prompt 5 - E-Library Module

```
Build the E-Library module.

Database model: library_books
Fields:
- id
- school_id
- title
- author
- description
- category
- subject
- reading_level
- language
- file_url
```

```
- cover_image_url
- uploaded_by
- created_at
- updated_at

Pages:
- /dashboard/library
- /dashboard/library/new
- /dashboard/library/[id]

Features:
- Upload PDF/book file
- Upload cover image
- Search books
- Filter by subject, language, reading level, category
- View/download book
- Teachers and admins can upload
- Students can view assigned school books only

Security:
- Enforce school_id on all queries.
- Prevent cross-school file access.
- Validate file types and size.
```

Prompt 6 - Video Lessons Module

```
Build the Video Lessons module.

Database models:
1. video_lessons
Fields:
- id
- school_id
- class_id
- subject_id
- title
- description
- video_url
- video_provider
- thumbnail_url
- duration_minutes
- visibility
- uploaded_by
- created_at
- updated_at

2. video_progress
Fields:
- id
- school_id
- video_lesson_id
- student_id
- watched_seconds
- completed
- last_watched_at

Pages:
- /dashboard/videos
- /dashboard/videos/new
- /dashboard/videos/[id]

Features:
- YouTube/Vimeo/private video URL support
- Class-based video access
- Subject filter
```

```
- Playlist-style lesson view
- Student progress tracking
- Teacher can see who watched the lesson

Security:
- Students can only access videos from their own school and assigned class.
- Teachers can only manage videos for their school/classes.
```

Prompt 7 - LMS, Assignments, Exams, and Grades

```
Build the LMS, Assignments, Exams, and Grades modules.

Features:
1. Subjects
2. Lessons
3. Lesson file uploads
4. Assignments
5. Student submissions
6. Teacher grading
7. Create exam
8. Link exam to class and subject
9. Add marks per student
10. Grade calculation
11. GED-style progress tracking
12. Student report card
13. PDF report export
14. Parent-friendly progress summary

Include GED subjects:
- Reasoning Through Language Arts
- Mathematical Reasoning
- Science
- Social Studies

Also support custom subjects.

UI:
- Lesson list
- Lesson detail
- Assignment detail
- Submission status
- Teacher grading table
- Student grade dashboard.
```

Prompt 8 - School Branding and Custom School Names

```
Build Custom School Branding.

Update schools table/model with:
- name
- short_name
- logo_url
- primary_color
- secondary_color
- address
- phone
- email
- website
- custom_domain
- subdomain

Build page:
- /dashboard/settings/branding
```

```
Features:
- Change school display name
- Upload logo
- Set theme colors
- Set address and contact
- Preview school header
- Apply branding to dashboard, reports, login page, and PDF exports
- Use the school name dynamically throughout the app

Security:
- Only school admins can edit branding for their own school.
- Super admin can edit any school branding.
```

Prompt 9 - Refugee Support Module

```
Build Refugee Support module.

Features:
1. Case notes per student
2. Sensitive note visibility control
3. Guardian/family situation notes
4. Sponsor support records
5. Document expiry reminders
6. Emergency contact panel
7. Referral tracking

Security:
- Case notes must be restricted.
- Teachers can only add basic notes.
- Only School Admin and approved Case Manager roles can see sensitive notes.
- Add audit logs for viewing or editing sensitive records.
- Never log sensitive note content.
- Add tests to ensure unauthorized users cannot access sensitive notes.
```

Prompt 10 - SaaS Multi-School Upgrade

```
Convert Refugee SchoolOS into a SaaS-ready multi-school platform.

Features:
1. Super admin dashboard
2. Create new school
3. School settings
4. School logo
5. School admin creation
6. Tenant isolation by school_id
7. Middleware to enforce school access
8. Prevent cross-school data leaks
9. Optional subdomain structure:
   - mrlc.refugeeschoolos.com
   - schoolname.refugeeschoolos.com

Security:
- Add tenant guard helper.
- Add automated tests for cross-school access prevention.
- Review every query and ensure school_id filtering is present.
```

Prompt 11 - Production Hardening

```
Perform a senior-level production hardening review and implementation.

Add:
1. Audit logs
2. Rate limiting
```

```
3. Input validation with Zod
4. Error handling
5. Database backup script
6. Environment variable validation
7. Secure headers
8. File upload validation
9. Role permission tests
10. Tenant isolation tests
11. Sensitive-data logging prevention

Also produce:
- README.md
- .env.example
- deployment guide for DigitalOcean + Nginx
- backup and restore guide
- basic troubleshooting guide.
```

7. Recommended Build Order

1. Project foundation, Prisma schema, auth, roles, and dashboard layout
2. Landing page and school login
3. Student records and guardian/emergency contacts
4. Classes, subjects, and teacher assignments
5. Attendance and reports
6. E-library
7. Video lessons and progress tracking
8. LMS assignments, exams, and grades
9. School branding and custom school names
10. Refugee support and sensitive permissions
11. SaaS multi-school tenant isolation
12. Production hardening and deployment